

JASON JOSHUA BEGARI

Email: bjjoshua122@gmail.com | Phone: (+91)-9494943567 | Location: Hyderabad, India

PROFESSIONAL SUMMARY

Electrical & Electronics Engineer with 3+ years in automation engineering, PLC programming, calibration systems, and R&D testing. Expert in Python automation with Modbus/RS485/RS232 communication, ISO 17025/4064/4185 compliance, test bench design, and NPD performance evaluation. Proven track record delivering 30% efficiency improvements and process optimization in industrial automation.

CORE SKILLS

Automation & Programming: PLC Programming (Siemens, Delta), VFDs, Python Automation, Modbus/RS485/RS232 Communication, Industrial Control Systems, Test Bench Design, Process Automation

Engineering & Calibration: ISO 17025/4064/4185 Compliance, NABL Laboratory Operations, Calibration Engineering, Equipment Validation, Medical Device Maintenance

R&D & Testing: NPD Performance Testing, Prototype Evaluation, Test Data Analysis, Design Validation, Metrological Compliance

Technical Tools: MATLAB Simulink, Circuit Design, Backend Development, System Optimization, Quality Control

PROFESSIONAL EXPERIENCE

R&D Test Engineer - Metrology | Kritsnam Technologies

July 2024 – Present

- Conducted New Product Development (NPD) testing and performance analysis of mechanical casings for ultrasonic water meters on custom-designed test benches
- Evaluated casing and meter prototypes against ISO 4064 performance criteria including accuracy, pressure loss, and repeatability across multiple flow rate conditions
- Analyzed experimental test data to identify design improvements, providing actionable feedback to the product design team for iterative casing development
- Collaborated with cross-functional teams to validate production-readiness of new meter designs, ensuring compliance with metrological and hydraulic standards

Automation and Calibration Engineer | Kritsnam Technologies

December 2023 – Present

- Automated mass production procedures using Python with Modbus/RS485/RS232 and Siemens & Delta PLC programming. This eliminated manual errors and improved efficiency by 30%
- Implemented ISO 4185 gravimetric calibration test bench across control electronics, pumping, and hydraulic systems
- Operated ISO 17025 compliant NABL-accredited Water Flow Meter Calibration Laboratory following ISO 4064 standards ensuring regulatory compliance
- Developed Python-based automation solution with ISO 4064/4185 compliance and Modbus/RS485/RS232 communication targeting 30% calibration time reduction and 150% daily output increase

Service Engineer | Techno Health Care Systems

September 2022 – October 2023

- Maintained critical medical equipment including ventilators, ECG machines, and patient monitors ensuring 99%+ uptime
- Performed equipment lifecycle services with calibration and validation per medical device standards
- Troubleshoot complex technical issues minimizing equipment downtime and maintaining operational reliability

Software Engineer Intern | Enquero Global LLP

August 2021 – July 2022

- Developed Python backend solutions for Symphony Messenger platform improving system stability
- Resolved security vulnerabilities and optimized system performance through code reviews and improvements

EDUCATION

B.Tech - Electrical & Electronics Engineering | JNTU Hyderabad | 2021 | CGPA: 7.25

KEY PROJECTS

Design and Performance Analysis of W-Pattern based Ultrasonic Water Flow Meter Casing (Current)

Design improvements for the new Dhaara Pulse W-Pattern based Ultrasonic Water Flow Meter Casing. Corrections based on performance analysis across multiple factors of accuracy and precision such as repeatability, positional reproducibility, orientational reproducibility, angular reproducibility with variable levels of flow turbulence based on International Standards OIML R49 and ISO4064.

Automated Water Meter Calibration System (Current)

Designed Python-based automation with Modbus/RS485/RS232 communication and ISO 4064/4185 compliance to reduce manpower from 2 engineers to 1 observer; targeting 30% time reduction and 150% output increase

DC Microgrid Protection Scheme (B.Tech Project)

Developed MATLAB simulation for disaster management with microsecond fault detection using Power MOSFETs and automated protection switching

Asian E-Bike Challenge (National Competition)

Engineered complete electric vehicle powertrain with 3kW/6.2kW BLDC motor, 48V lithium-ion battery, and custom control electronics